# $CO_2$-EOR DIGITAL TWIN

Delaware Basin $CO_2$ Enhanced Oil Recovery Command Center

DATABRICKS APPS

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## NEAR REAL-TIME DIGITAL TWIN & MULTI-AGENT AI FOR $CO_2$-EOR FIELD OPERATIONS

$CO_2$-EOR Digital Twin delivers a unified command center fusing geospatial field monitoring, subsurface fiber optic sensing (DAS/DTS), multi-agent AI advisory, injection pattern optimization, and carbon accounting on the Databricks Lakehouse Platform.

### ◆ BUSINESS VALUE

- **Maximize oil recovery** with AI-optimized $CO_2$ injection patterns and WAG ratio tuning
- **Track carbon storage compliance** with real-time $CO_2$ balance and accounting dashboards
- **Reduce subsurface uncertainty** with integrated DAS/DTS fiber optic and microseismic monitoring
- **Improve field economics** with per-well revenue, OpEx, and netback analysis
- **Accelerate decision-making** through 5 specialized AI agents providing contextual recommendations

### ◆ CHALLENGES ADDRESSED

- **Fragmented field data** — SCADA, reservoir models, and economic tools in disconnected systems
- **Limited subsurface visibility** — no integrated view of DAS, DTS, and microseismic data
- **Manual carbon accounting** — spreadsheet-based $CO_2$ tracking prone to errors
- **Reactive pattern management** — injection adjustments based on lagging monthly reports
- **Siloed expertise** — reservoir, operations, and commercial teams working independently

### ◆ WHO IT'S FOR

🛢️ **Reservoir Engineer**
Subsurface monitoring, saturation maps, pressure analysis, and pattern optimization

📊 **Operations Manager**
Real-time field overview, well status, and operational KPI tracking

🌱 **Carbon/ESG Analyst**
$CO_2$ balance tracking, storage compliance, and emissions reporting

💰 **Commercial Analyst**
Per-well economics, netback analysis, and revenue forecasting

| 24 | 5 | 8 | 4 |
|---|---|---|---|
| Wells Monitored | AI Agents | Dashboard Tabs | Flood Patterns |

### ◆ QUICK START GUIDE

1. **Clone Repository**
   Clone the $CO_2$-EOR Digital Twin repo to your local machine
2. **Build Frontend**
   Run npm install & npm run build for the React UI
3. **Deploy the App**
   Run databricks apps deploy to launch the command center
4. **Monitor & Explore**
   Open the app URL and explore all 8 tabs

### ◆ GETTING THE MOST OUT OF CO2-EOR DIGITAL TWIN

📡 **Explore the Field Map**
Use the MapLibre GL geospatial overview to monitor wells, pipelines, facilities, and fleet vehicles

🤖 **Consult AI Agents**
Chat with 5 specialized agents for monitoring, optimization, maintenance, and commercial insights

🔬 **Dive into Subsurface**
Analyze DAS waterfall plots, DTS profiles, reservoir saturation maps, and microseismic events

⚖️ **Track Carbon Balance**
Monitor injected, recycled, purchased, and stored $CO_2$ volumes for compliance

### ◆ TARGET INDUSTRY

**Upstream Oil & Gas** — $CO_2$ Enhanced Oil Recovery operations, mature field rejuvenation, carbon capture utilization & storage (CCUS), and tertiary recovery programs in the Permian and Delaware Basins.

⚠ **DEMO ONLY** — This application is a demonstration Databricks App. Please refer to the included LICENSE file for full terms of use.

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